Mojo Programming Language - 4 Module Syllabus

MODULE I – Introduction to Mojo Programming

• Introduction: What is Mojo, History and evolution, Why Mojo was created, Features, Advantages and limitations, Applications.

• Installation and Setup: Installing Mojo SDK, VS Code setup, First program, Mojo REPL.

• Basic Syntax: Program structure, Comments, Variables, Constants, Data Types, Input/Output, Operators, Type Conversion.

• Control Flow: if, else, elif, match, for, while, break, continue, pass.

• Lab: Hello World, Calculator, Largest of three numbers, Multiplication table, Prime checker.

MODULE II – Functions and Data Structures

• Functions: Defining, Calling, Parameters, Return values, Default/Keyword arguments, Recursion, Variable scope.

• Collections: Lists, Tuples, Dictionaries, Sets, Strings.

• String Handling: Operations, Methods, Formatting, f-strings.

• Exception Handling: try, except, else, finally, raise.

• Lab: Grade calculator, Fibonacci, Factorial, Phone book, Palindrome checker.

MODULE III – Object-Oriented Programming and Advanced Mojo

• OOP: Classes, Objects, Constructors, Instance variables, Methods, Encapsulation.

• Inheritance: Single, Multiple, Overriding, super().

• Polymorphism: Overriding, Duck typing, Overloading concept.

• Mojo Features: Structs, Traits, Ownership, Memory safety, Value semantics, SIMD basics, Compile-time features.

• Lab: Employee, Library, Bank Account, Student Record, Shape Area Calculator.

MODULE IV – High Performance Computing and AI with Mojo

• Performance: Python interoperability, Import Python libraries, Performance comparison, Optimized Mojo code.

• Parallel & Vector Computing: SIMD, Parallel execution basics, Benchmarking.

• AI & ML: Tensor operations, Numerical computing, MAX Engine, AI workflow.

• Mini Project: Planning, Organization, Debugging, Testing, Documentation, Deployment.

• Lab: Matrix multiplication, Python vs Mojo benchmark, Image processing, Numerical computation, AI mini project.

Course Outcomes

• Understand Mojo fundamentals and syntax.

• Develop programs using functions, collections, and exception handling.

• Design applications using OOP and Mojo-specific features.

• Build high-performance and AI-oriented applications using Mojo.